Digital Receipt Vault using Blockchain & IPFS

A PROJECT REPORT

Submitted by

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ABSTRACT

The growing use of digital receipts and documents has highlighted major challenges such as forgery, centralized control, and unreliable verification. Conventional systems rely on centralized databases, making documents vulnerable to tampering, data breaches, and single points of failure. To overcome these limitations, this project presents a **decentralized and cryptographically secure digital receipt issuance and verification system** using **IPFS, blockchain technology, and public-key cryptography**.

In the proposed system, receipt files are uploaded by the issuer and stored in **IPFS**, which generates a unique **Content Identifier (CID)** based on the file's cryptographic hash, ensuring data integrity and immutability. The issuer digitally signs the CID along with receipt metadata and generates a claimable link for the user. The user claims ownership by signing the receipt using their private key, and the signed transaction is permanently recorded on the **Ethereum blockchain** via a **Solidity smart contract**. The smart contract stores only essential metadata such as the CID, issuer and owner addresses, signatures, and timestamps, ensuring scalability and low on-chain storage cost.

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LIST OF ABBREVIATIONS

IPFS	_	InterPlanetary File System
CID	_	Content Identifier
SHA	_	Secure Hash Algorithm
PKI	_	Public Key Infrastructure
ECDSA	_	Elliptic Curve Digital Signature Algorithm
ETH	_	Ethereum
IDE	_	Integrated Development Environment
SMTP	_	Simple Mail Transfer Protocol
P2P	_	Peer-to-Peer
EVM	_	Ethereum Virtual machine

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The rapid digitization of financial and official transactions has led to widespread use of digital receipts and documents. While digital formats improve accessibility and efficiency, they also introduce challenges related to security, authenticity, and trust. Traditional systems rely heavily on centralized databases and authorities for document issuance and verification, which makes them vulnerable to tampering, data breaches, and unauthorized access.

Blockchain technology, combined with decentralized storage and cryptographic techniques, offers a promising solution to address these issues by enabling immutable, transparent, and trustless document verification systems. To overcome these limitations, this project presents a **decentralized and cryptographically secure digital receipt issuance and verification system** using **IPFS, blockchain technology, and public-key cryptography**.

In the proposed system, receipt files are uploaded by the issuer and stored in **IPFS**, which generates a unique **Content Identifier (CID)** based on the file's cryptographic hash, ensuring data integrity and immutability. The issuer digitally signs the CID along with receipt metadata and generates a claimable link for the user. The user claims ownership by signing the receipt using their private key, and the signed transaction is permanently recorded on the **Ethereum blockchain** via a **Solidity smart contract**.

1.2 PROBLEM STATEMENT

- Existing digital document management systems lack a reliable mechanism to guarantee document authenticity, immutability, and ownership.
- Centralized storage makes documents susceptible to tampering and unauthorized access. Verification depends on the issuing organization, resulting in delayed and unreliable third-party validation.
- There is no standardized, tamper-proof method to prove that a document was issued by a legitimate authority and has not been altered over time.
- There is a need for a decentralized and cryptographically secure system for document issuance and verification.

1.3 OBJECTIVES

- To design and implement a **decentralized digital receipt and document verification system**.
- To ensure **immutability and integrity** of documents using blockchain and contentaddressed storage.
- To enable **cryptographic proof of ownership** through digital signatures.
- To support **trustless third-party verification** without dependence on centralized authorities.
- To provide a **scalable and cost-efficient** solution by storing only essential metadata on-chain.

1.4 OVERVIEW OF THE PROPOSED SOLUTION

The proposed solution follows a layered architecture consisting of application, backend, storage, and blockchain layers. Receipt files are uploaded by the issuer and stored in **IPFS**, which generates a unique **Content Identifier (CID)** based on the file's cryptographic hash. The issuer digitally signs the CID and receipt metadata, and a

claimable link is generated for the user. The user claims ownership by signing the receipt, and the transaction is recorded on the **Ethereum blockchain** using a **Solidity smart contract**. The smart contract stores metadata such as the CID, issuer address, owner address, cryptographic signatures, and timestamps. Verification is performed by deriving public keys from stored signatures, enabling secure and trustless validation.

1.5 SCOPE OF THE PROJECT

- Applicable to **digital receipts, invoices, certificates, and official documents**.
- Supports **public verification** without revealing sensitive document content.
- Eliminates document forgery and unauthorized modification through immutability.
- Can be extended to support **NFT-based ownership**, multi-signature verification, or privacy-enhancing techniques.
- Suitable for use in **finance, education, healthcare, legal systems, and egovernance** platforms.

CHAPTER 2

LITERATURE SURVEY

Key Findings from Literature Survey

- **Adoption of Hybrid Blockchain–IPFS Architectures**

  The literature consistently shows that hybrid architectures combining blockchain and IPFS are preferred over fully on-chain storage systems. Large data such as documents, receipts, and records are stored off-chain in IPFS, while blockchain is used to store hashes and metadata for integrity verification. This approach significantly reduces storage cost, improves scalability, and maintains immutability, making it suitable for real-world applications involving large volumes of data.

- **Extensive Use of Smart Contracts for Trust Automation**

  Smart contracts are widely utilized to automate processes such as receipt issuance, verification, access control, and fair exchange. By embedding business logic directly into smart contracts, these systems eliminate dependency on centralized authorities and trusted third parties. The reviewed studies demonstrate that smart contracts ensure transparency, enforce predefined rules, and provide non-repudiation in digital transactions.

- **Strong Emphasis on Cryptographic Security Mechanisms**

  Security is a key focus in all reviewed papers, with extensive use of cryptographic techniques including digital signatures, hashing, encryption algorithms, and key management schemes. These mechanisms ensure data authenticity, confidentiality, and integrity throughout the system. The integration of cryptography with blockchain immutability effectively prevents data tampering, forgery, and unauthorized access.

- **Emerging Focus on Scalability and Practical Deployment**

  Recent research increasingly addresses performance limitations of blockchain through off-chain computation, middleware layers, and cross-chain mechanisms. These optimizations improve transaction throughput and reduce latency, enabling deployment in complex scenarios such as supply chains and multi-party contracts. However, the literature also highlights the need for simpler architectures and better user experience to support wider adoption.

Title	Authors	Year and Mode of Publication	Methodology used
An integrated blockchain and IPFS-based solution for secure and efficient source code repository hosting using middleman approach	Md. Rafid Haque, Sakibul Islam Munna, Sabbir Ahmed	2025 – Arxiv	Hybrid architecture using IPFS for off-chain storage, Ethereum smart contracts for hash verification, and a middleware layer for encryption and CID management.
A Security-Oriented Data-Sharing Scheme Based on Blockchain	Wei Ma	2024 – MDPI	Blockchain-based data sharing using smart contracts, ECC encryption, and digital signatures to enforce access control and data integrity.
A cross-chain model for warehouse receipts in port supply chain based on notary mechanism and ShangMi cryptographic algorithms	Yangbo Chen, Wei Ou	2025 - ResearchGate	Cross-chain blockchain model with notary mechanism and ShangMi cryptographic algorithms for secure receipt verification and transfer.

Title	Authors	Year and Mode of Publication	Methodology used
A Blockchain-Enabled Framework for Storage and Retrieval of Social Data	Aishwarya Parab , Prakhar Pradha, Yogesh Simmhan	2025 - Arxiv	Hybrid IPFS–blockchain framework with smart contracts for data provenance, access control, and trust management.
A Blockchain-Based Protocol for Fair Delivery for Receipts	M. Francisca Hinarejos, Josep-Lluis Ferrer-Gomila	2024 - MDPI	Smart contract–based fair exchange protocol using cryptographic commitments and digital signatures to ensure non-repudiation.
A Fair Multi-Party Contract Signing Scheme Based on Off-Chain Protocols and On-Chain Smart Contracts	Rong Jiang Yulin Li Xuetao Pu Xueke Wang	2025 - Springer	Off-chain signature exchange combined with on-chain smart contract verification and penalty enforcement.

CHAPTER 3

SYSTEMARCHITECTURE

3.1 System Overview :

Storage Layer (IPFS) IPFS (InterPlanetary File System)

- Stores receipt files in a **decentralized and distributed manner**.
- Generates a **Content Identifier (CID)** based on the cryptographic hash of the file.
- Ensures **immutability**, as any change in the file results in a new CID.
- Prevents unauthorized modification of stored documents.

Only the CID is used for blockchain reference, reducing storage overhead.

Blockchain Layer Ethereum Blockchain

- Provides an immutable ledger to store receipt verification data.
- Ensures transparency, security, and tamper resistance.

Solidity Smart Contract

Stores only essential metadata:

1. Content Identifier (CID)
2. Receipt Hash
3. Issuer's Digital Signature
4. Issuer's Blockchain Address
5. Owner's Blockchain Address
6. Issuance Timestamp

Smart Contract Responsibilities

- Records receipt issuance and ownership claims.
- Verifies cryptographic signatures.
- Enables public, trustless verification using CID.
- Prevents duplicate claims or unauthorized ownership changes

Third-Party Verification Component

- Any external entity (employer, auditor, institution) can verify a receipt.
- Verification is done by submitting the CID to the smart contract.
- Public keys are derived from stored signatures to validate:
  - Authenticity of the issuer
  - Ownership of the user
- No interaction with the issuing company is required.
- Integrated PostgreSQL for structured data management, including user profiles, products, and order histories.

Upload Request Module

- Receives receipt files from the company page , Uploads files to IPFS.
- Retrieves the generated CID and sends it back to the application layer.

Claim Link Generation

- Creates a secure, unique claimable link associated with the CID.
- Ensures only the intended recipient can claim ownership.

Notification / Email Service

- Sends the claimable link to the user via email.
- Ensures secure delivery of receipt ownership requests.

The backend does not store documents permanently, maintaining decentralization.

User Interaction Interfaces

- Provides separate interfaces for issuers, recipients, and verifiers to interact with the system securely.
- Enables issuing organizations to upload digital receipts, obtain the IPFS-generated **Content Identifier (CID)**, digitally sign the receipt metadata, and generate a secure claimable link for recipients.
- Allows recipients to access the claimable link, cryptographically sign the receipt using their private key, and claim ownership by submitting the signed transaction to the blockchain.

- Supports public, read-only verification where third parties can validate receipt authenticity using the CID without accessing or revealing sensitive document content

3.6 ARCHITECTURE DIAGRAM

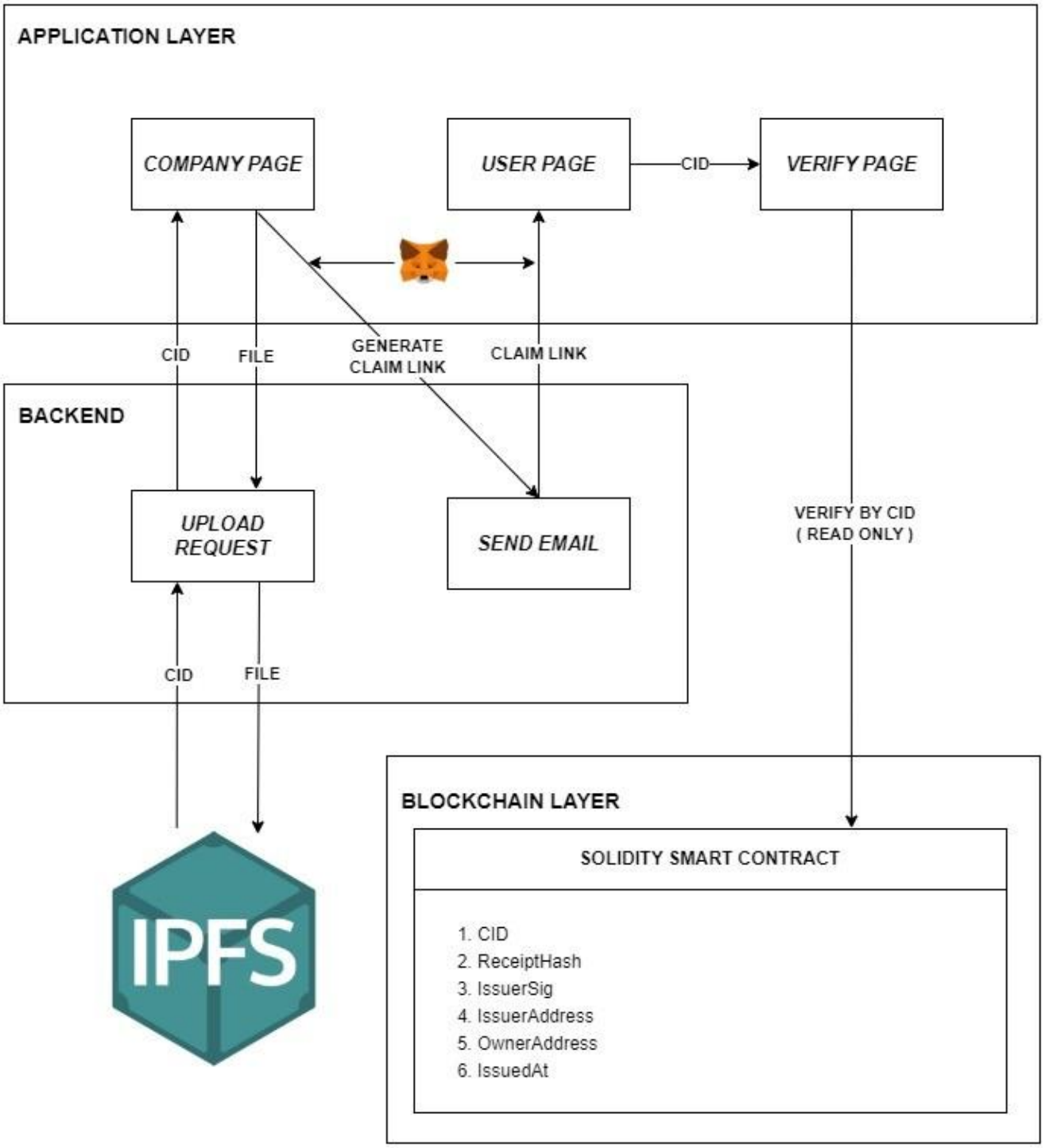


Fig 3.1 System Architecture

Cryptographic + Blockchain flow :

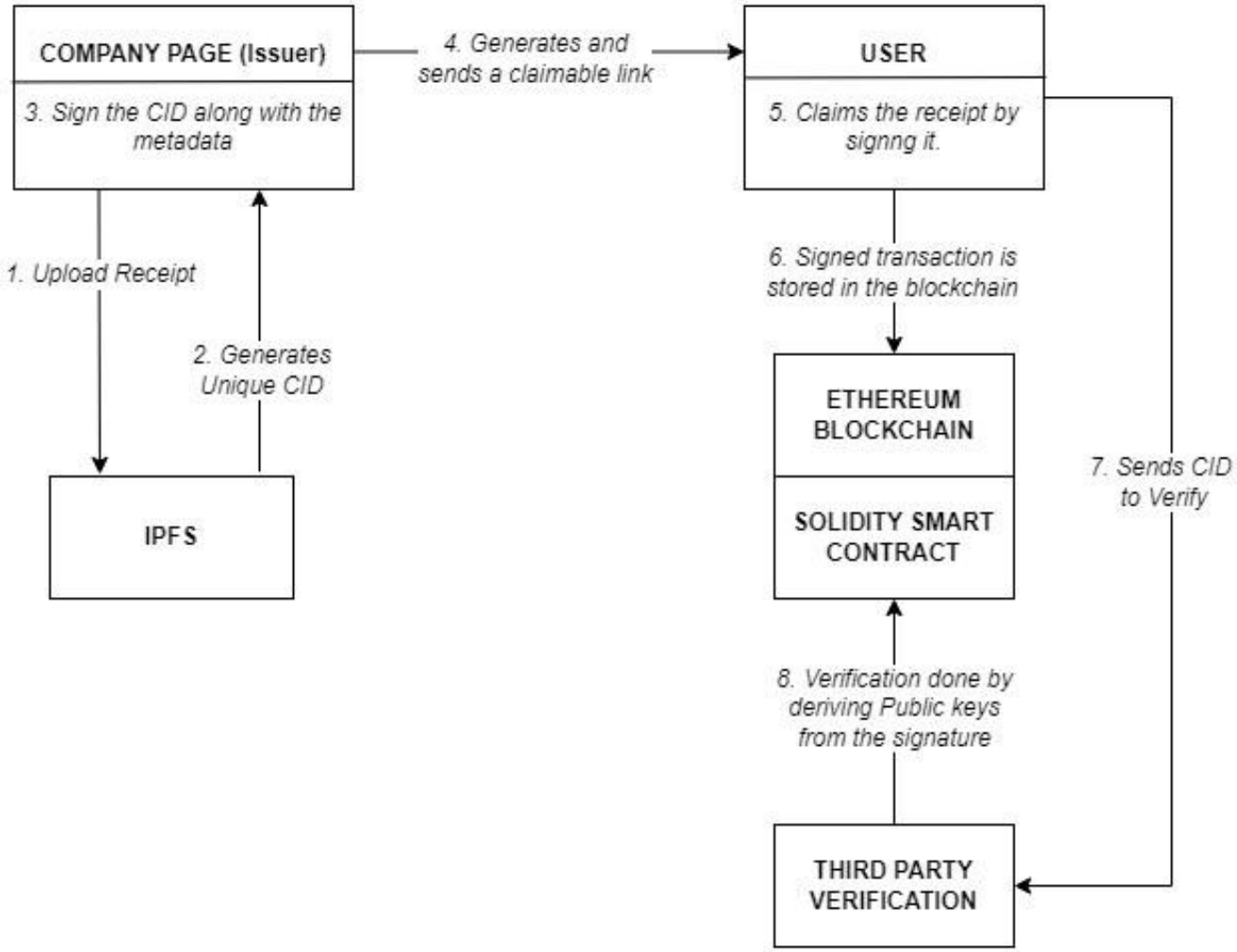


Fig 3.2 Cryptographic + Blockchain Flow

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

- **Frontend Technologies**

  HTML, CSS, JavaScript with React.js for building issuer, recipient, and verification interfaces.

- **Backend Technologies**

  Node.js with Express.js for handling API requests, IPFS interaction, claim-link generation, and email services.

- **Blockchain Platform**

  Ethereum blockchain (Testnet - Sepolia).

- **Smart Contract Language**

  Solidity for implementing receipt issuance, ownership claiming, and verification logic.

- **Blockchain Development Tools**

  Hardhat for smart contract development, testing, and deployment.

- **Wallet & Web3 Integration**

  MetaMask for user authentication and transaction signing.

- **Decentralized Storage**

  IPFS (via Lighthouse SDK).

CHAPTER 5
SYSTEM DESIGN AND IMPLEMENTATION

5.1 MODULES COMPLETED

1. Issuer Module

- Provides an interface for organizations to upload digital receipts or documents.
- Handles receipt issuance and initiates the IPFS upload process.
- Displays the generated Content Identifier (CID) returned from IPFS.
- Digitally signs the CID along with receipt metadata to ensure authenticity.
- Generates a secure claimable link for the intended recipient.

2. IPFS Storage Module

- Stores uploaded receipt files in a decentralized and distributed manner.
- Generates a unique CID based on the cryptographic hash of the document.
- Ensures immutability, as any modification to the file results in a new CID.
- Reduces on-chain storage by keeping documents off the blockchain.

3. Recipient (User) Module

- Allows users to access the receipt using the claimable link.
- Enables users to cryptographically sign the receipt using their private key.
- Confirms receipt ownership by submitting the signed claim to the blockchain.
- Prevents unauthorized ownership claims.

4. Smart Contract Module

- Implements receipt issuance, ownership claiming, and verification logic using Solidity.
- Stores essential metadata such as CID, receipt hash, issuer signature, issuer address, owner address, and issuance timestamp.
- Ensures immutability and transparency of receipt records.
- Prevents duplicate or fraudulent receipt claims.

5. Verification Module

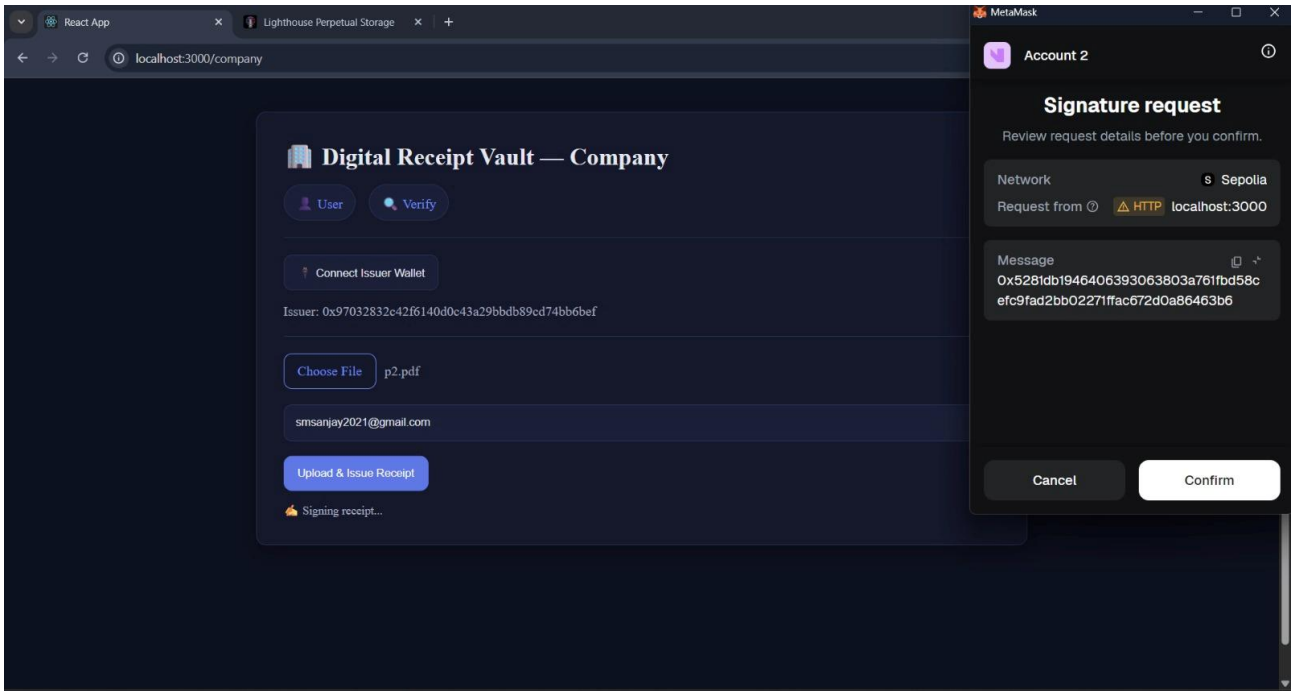
- Allows third parties to verify receipts using the CID.
- Retrieves and validates blockchain data in read-only mode.
- Verifies issuer and owner authenticity using cryptographic signatures.
- Does not expose sensitive document content during verification.

6. Backend Services Module

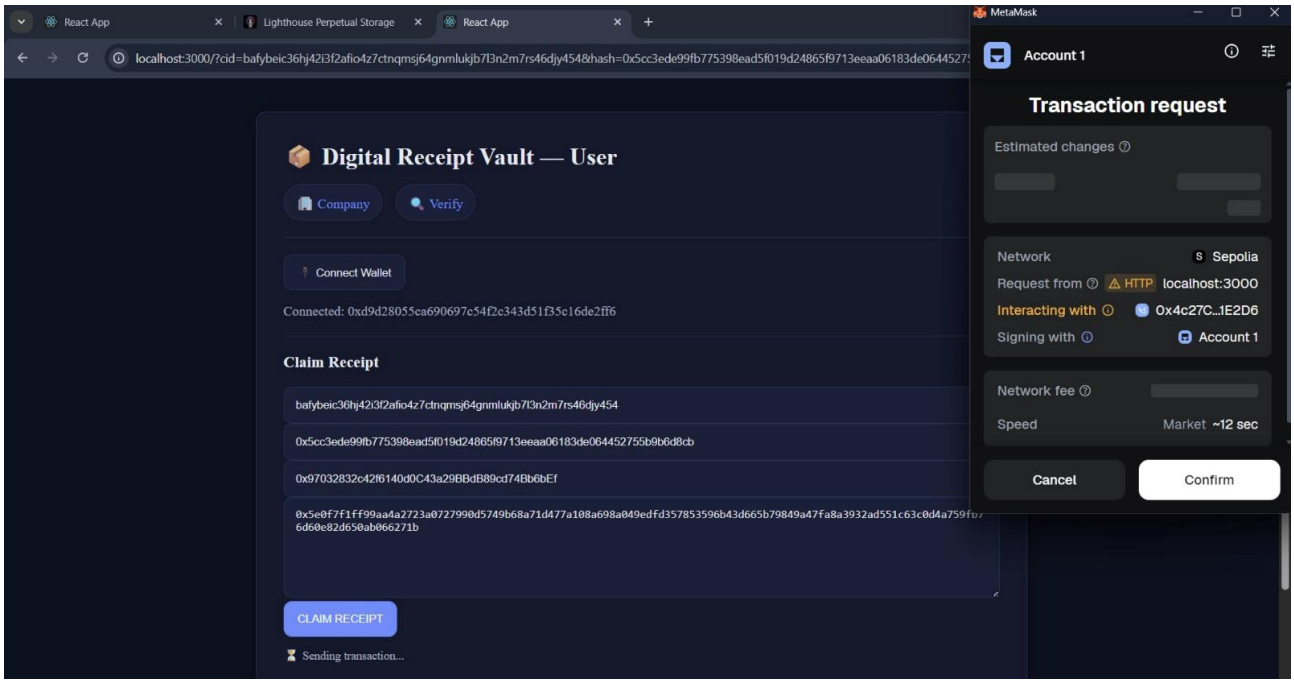
- Acts as an intermediary between frontend, IPFS, and blockchain layers.
- Handles file upload requests and IPFS interactions.
- Generates claimable links and manages email notifications.
- Ensures secure API communication.

CHAPTER 6
RESULTS AND DISCUSSION

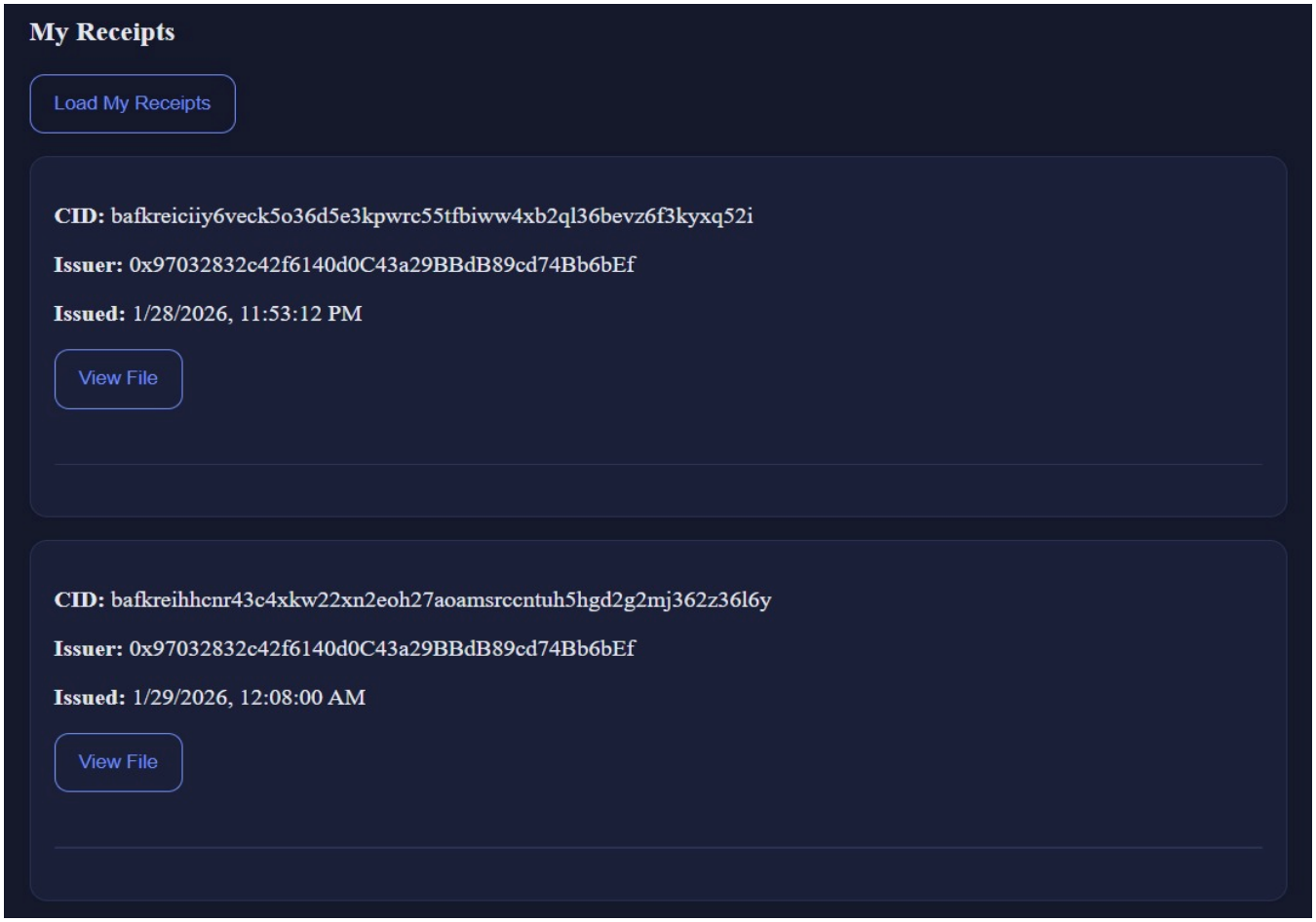
6.1 Company Page



6.2 User Page



6.3 User Receipts



6.4 Claim Link

6.5 IPFS Page

Home

Search Name / CID

Filter

Name	CID	Size	File Type	Last Modified
blob	bafybeihi2f6iqyqhada2oucaczmnkhaukqwejgoj6x7zfyjcdzvwtdxrem	3.89 MB	Public	10-02-2026
blob	bafybeic36hj42i3f2afio4z7ctnqmsj64gnmlukjb7l3n2m7rs46djy454	2.63 MB	Public	10-02-2026

6.6 Receipt Verification

Verify Receipt (Public)

User Company

bafkreiciiy6veck5o36d5e3kpwrc55tfbiww4xb2ql36bevz6f3kyxq52i

Verify Receipt

Receipt verified successfully

VALID RECEIPT

Issuer: 0x97032832c42f6140d0C43a29BBdB89cd74Bb6bEf

Owner: 0xD9D28055CA690697c54f2C343D51f35c16De2FF6

Issued At: 1/28/2026, 11:53:12 PM

CHAPTER 7

CONCLUSION :

This project successfully demonstrates a **decentralized, secure, and tamperproof digital receipt verification system** by integrating **IPFS, blockchain technology, and cryptographic signatures**. By eliminating reliance on centralized authorities, the system ensures document authenticity, integrity, and transparent ownership through immutable blockchain records and contentaddressed storage.

The use of **IPFS** enables efficient and decentralized storage of receipt files, while the **Ethereum blockchain and Solidity smart contracts** provide a permanent and verifiable ledger for receipt metadata and ownership claims. The claim-based ownership mechanism and trustless third-party verification ensure that receipts can be securely issued, claimed, and validated without manual intervention.

Overall, the proposed solution offers a scalable and reliable framework for digital document management, with strong potential for adoption in financial, educational, legal, and e-governance applications.

CHAPTER 8

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